

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2461067.8102 +/- 0.0028 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.048 +/- 0.057
Transit depth (Rp/Rs)^2	0.23 +/- 0.55 [%]
Orbital Inclination (inc)	77.49 +/- 2.95
Airmass coefficient 1 (a1)	0.58 +/- 0.31
Airmass coefficient 2 (a2)	0.46 +/- 0.3
Scatter in the residuals of the lightcurve fit is	50.03 %
Best Comparison Star	None
Optimal Aperture	1.22
Optimal Annulus	3.0
Transit Duration (day)	0.017 +/- 0.018

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R fit vs published	0.048 ± 0.057 vs 0.1594 (deviation 1.65σ)
Duration fit vs published	0.017 d vs 0.0512 d (deviation 1.6σ)
Mid-time O-C	-4.9 min
Depth SNR	0.7
Residual scatter	50.03 %

Light curve

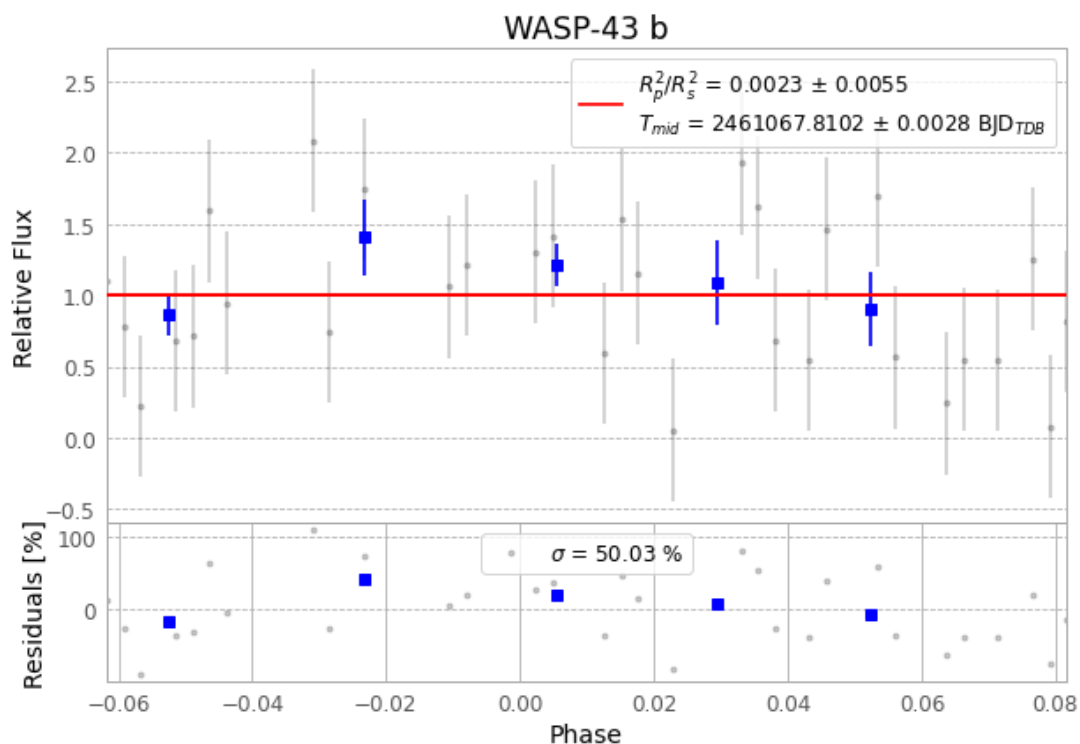


Figure 1 — FinalLightCurve_WASP-43 b_2026-01-26.png (SHA-256 e89652465748cde3...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [182, 379], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 c1942a372e41a069...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 8b944da

Data provenance

59 FITS frames (39 MB), WASP-43260127060615.FITS → WASP-43260127090015.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-43-260127.log (a2e0ba4fde6c...), FinalLightCurve_WASP-43 b_2026-01-26.png (e89652465748...), FinalLightCurve_WASP-43 b_2026-01-26.csv (39a6cae21326...)

Compute resources

Wall time 175.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-18 11:14Z.